

The asset allocation that maximizes the sustainable withdrawal rate of 4.4 percent is identified with a circle. In addition to the optimal asset allocation, the range of points on the efficient frontier (representing a range of asset allocations) that allow for a withdrawal rate within 0.1 percentage points of the maximum are highlighted with a thicker line.

But what *is* the optimal asset allocation, and what range of asset allocations perform nearly as well as the optimal allocation? Though the exhibit does not allow us to see this directly, these answers are provided more clearly in Exhibit 4.8, which shows results for a wide variety of retirement durations and failure rates in the same sort of way as the Trinity study. For the case just discussed—a thirty-year retirement with a 10 percent acceptable failure rate—the third column indicates a maximum sustainable withdrawal rate of 4.4 percent with these conditions. The next set of columns shows that the optimal asset allocation to achieve this 4.4 percent withdrawal rate is 45 percent stocks and 55 percent bonds. This is a fixed asset allocation held throughout retirement with annual rebalancing.

The last four columns demonstrate the range of allocations in which one could expect to do nearly as well with a withdrawal rate within 0.1 percent of the maximum. Any stock allocation between 34 and 71 percent would support a withdrawal rate of 4.3 percent.

The exhibit further shows, unsurprisingly, that sustainable withdrawal rates are higher for shorter retirement durations and higher allowable failure probabilities. In addition, the optimal stock allocation tends to increase both for longer retirement durations and higher allowed failure probabilities. This suggests that the upside potential from stocks is increasingly important to sustaining a longer retirement, and that an acceptance of greater lifestyle risk (the risk of portfolio depletion) allows for more aggressive behavior in relation to a higher spending rate and stock allocation.

Exhibit 4.8

Withdrawal Rate and Asset Allocation Guidelines for Retirees

Inflation-Adjusted Withdrawals